



Unit 3 · Outcome 2 · Sustainable Development

Who decides how much water the wetland gets?

KK06

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



THE FOUR FACTORS

02 A decision sits at the centre of a web

Every environmental decision at Boorla is pushed and pulled by the same four factors. Drag to orbit. Click a factor to light up its links to the others.

VCAA names exactly four factors. Memorise them as S·R·D·T — and notice that two are about people and two are about evidence.

What different groups care about, what they know (including Traditional knowledge), and what they want most. Irrigators, the town, Traditional Owners and conservation managers all value the river differently.

The laws, treaties, plans and entitlements that set the rules — from the international Ramsar Convention down to a state water-sharing plan. They constrain what managers are allowed to do.

Historical and current evidence — flood records, fish recruitment surveys, vegetation condition, hydrological models. It tells managers what is happening and what might happen, but it carries uncertainty.

Tools that change what's possible — drip irrigation, soil-moisture sensors, satellite greenness mapping, fishways, real-time flow telemetry. Technology can ease a tension or create a new one.

◆ ANALOGY

Analogy · the mixing desk Picture a sound engineer at a mixing desk with four sliders. Push stakeholder demand up and the environmental flow slider has to drop — they're patched together. New technology can re-wire the desk so one slider no longer fights the other. "Responsible" doesn't mean every slider at maximum; it means a balance you can justify against the sustainability principles.

THE SPINE OF THE DOT POINT

03 Interconnections and tensions — learn the difference cold

The two words in the dot point are the whole game. An answer that only lists the four factors will sit on low marks. The marks live in showing how they relate.

Where two factors reinforce or depend on each other — pulling the same way. Example: new sensor technology generates the scientific data that regulatory targets are built on. Move one, the other moves with it.

Where two factors pull against each other — satisfying one strains the other. Example: irrigators' priority (maximum allocation) clashes with the Basin Plan's regulated environmental-flow target. A gain for one is a loss for the other.

• NOTE

Tensions can be within a factor, not just between The Boorla people and Connawarra's irrigators are both stakeholders — yet their priorities conflict. That's a tension within the single factor "stakeholder values". VCAA's wording is "within and between factors", so watch for both.

FACTOR S · THE PEOPLE

04 Four groups, one river

Priority: secure, affordable water for dairy and crops near Connawarra. Value system leans anthropocentric — the river's worth is its use to people. Hold water entitlements.

Priority: reliable, safe drinking water and local jobs. Worried about a drying river through town and the cost of water restrictions.

Priority: cultural flows that flood Country and sustain significant sites and species. Hold deep Traditional ecological knowledge of the wetland's natural flood rhythm — a value system closer to ecocentric.

Priority: the long-term ecological health of a Ramsar wetland — bird breeding, fish recruitment, river red gum survival. Acts on behalf of the environment; broadly ecocentric.

• **NOTE**

Link back to KK14 / value systems Stakeholder values are anchored in the value systems you met earlier — anthropocentric, biocentric, ecocentric, technocentric. Naming a group's value system is a fast way to earn an "analyse" mark, because it explains why their priority is what it is.

FACTOR R · THE RULES

05 Regulatory frameworks set the boundaries

Managers at Boorla cannot simply do what the science suggests or what one group wants. They operate inside a stack of binding rules — international, national, state and local. These frameworks inform and constrain the management strategy.

- Ramsar Convention (international) — Boorla is a listed wetland of international importance; Australia is obliged to maintain its "ecological character". This elevates the wetland's claim on water.
- Water Act 2007 (Cth) & the Murray–Darling Basin Plan — set a legal cap on extraction and a volume of water reserved for the environment, delivered by the Commonwealth Environmental Water Holder.
- EPBC Act 1999 (Cth) — protects nationally listed migratory waterbirds and threatened species that use the wetland.
- Victorian water law & entitlements — water-sharing arrangements that define how much each licensed irrigator may take.
- Native title & cultural water — emerging recognition of Traditional Owners' rights to water for cultural purposes.

⚠ **WATCH OUT**

Frameworks interconnect — and can clash with each other Ramsar and the Basin Plan reinforce the case for environmental flows (interconnection). But a state irrigation entitlement granted decades ago can conflict with a newer environmental target (tension). Examiners love this: a tension can exist between two regulations, not just between a rule and a person.

FACTOR D · THE EVIDENCE

06 Historical and current scientific data

VCAA is deliberate about the words "historical and current". Responsible decisions compare what the system used to do with what it is doing now.

Long-term flood records, old aerial photographs, fish and bird counts going back decades, and Traditional knowledge of the wetland's natural flood rhythm. It sets the baseline — what "healthy" looked like.

Real-time flow gauges, recent vegetation-condition surveys, water-quality probes and satellite greenness. It shows the present trajectory and whether a strategy is working.

Interpretation is where tensions appear. The same dataset can be read differently: an irrigator may point to a recent wet year as proof the wetland is "fine"; an ecologist reads the decadal trend and sees decline. Data also carries uncertainty — which is exactly why the precautionary principle matters.

• **NOTE**

The exam-grade move Don't write "scientists collected data". Write which data (e.g. river red gum canopy-condition surveys), what it showed (worsening stress below a flooding threshold), and how it was used (to justify a managed spring flood). Specificity is marks.

FACTOR T · THE TOOLS

07 New technologies — the double-edged factor

Technology can dissolve a tension or manufacture one. That double edge is what makes it examinable.

Drip irrigation & soil-moisture sensors let farms grow the same crop with less water — freeing volume for the wetland. Fishways let Murray cod pass the dam. Satellite greenness (NDVI) cheaply monitors recovery.

New tech costs money — who pays? (the user-pays principle). Efficient irrigation can encourage more planting (the "rebound effect"), erasing the water saved. A desalination plant for the town shifts the problem to energy and emissions.

◆ **ANALOGY**

Analogy · a brighter torch in a bigger cave Technology is like upgrading your torch. A brighter beam (better data, less waste) helps you see — but it also reveals how much cave you hadn't noticed, and someone has to pay for the batteries. New tools rarely give something for nothing; they shift the trade-off.

PUTTING IT TOGETHER

08 A worked decision: the spring environmental flow

The CMA proposes releasing a managed spring flood from Yarrin Dam to trigger river red gum growth and waterbird breeding. Watch all four factors switch on at once — some reinforcing, some resisting.

- Scientific data (D) shows canopy condition has crossed a stress threshold and birds haven't bred in three years — a trigger for action.
- Regulatory frameworks (R) make the flow possible: the Basin Plan reserves the environmental water, and Ramsar obliges Australia to protect ecological character. ↳ R interconnects with D.
- Stakeholder values (S) split: Traditional Owners and the CMA support the flow; some irrigators fear it signals less water for them next season. ↳ tension within S, and between S and R.
- New technology (T) defuses part of the conflict: soil-moisture sensors let farms cut on-farm use so the environmental flow doesn't come straight out of their allocation. ↳ T eases the S–R tension.

• **NOTE**

What "responsible" looks like here The decision is responsible not because everyone is happy, but because it is evidence-based (D), lawful (R), transparent about trade-offs (S), uses the best available tools (T), and can be justified against the sustainability principles — intergenerational equity, the precautionary principle and ecological integrity.

DECODE THE QUESTION

09 Command-word decoder

◆ COMMAND WORDS

The same content earns different marks depending on the command word. Tap a card to see what the examiner actually wants for this dot point.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

WHERE MARKS LEAK AWAY

10 Six named exam traps

△ EXAM TRAPS

Naming S, R, D, T earns recall marks only. Fix: for every factor, state one interconnection or tension with another. They're opposites. Fix: interconnection = reinforcing; tension = conflicting. Use both words explicitly. "The community" is vague. Fix: name the Boorla groups and their specific priority and value system. Students cast tech as a pure fix. Fix: note a cost, an equity issue or a rebound effect. Only mentioning present monitoring. Fix: contrast a baseline (historical) with a trajectory (current). This is Outcome 2 — principles are the yardstick. Fix: tie the decision to equity / precautionary / ecological integrity.

WRITE LIKE A STATE-RANKER

11 P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P-E-L WRITING

Structure every extended-response paragraph as Point → Evidence → Link. The prompt below is a typical 4-mark "analyse" task. State the tension in one sentence, naming the two factors. Anchor it in the case study with a specific, named example for each factor. Explain the consequence for the decision and tie it to a sustainability principle. A clear tension exists between stakeholder priorities and regulatory frameworks. Connawarra's irrigators prioritise maximum water allocation for dairy and horticulture, yet the Murray–Darling Basin Plan legally reserves a volume of environmental water for the Boorla Wetlands. Because both draw on the one regulated river, water delivered to the wetland reduces the volume available to irrigators in the same season. This means the responsible decision cannot satisfy both, and managers must privilege the Basin Plan's intergenerational-equity obligation over short-term economic priority — accepting a present cost to irrigators to protect the wetland for future generations.

PRACTICE & SELF-MARK · 20 MARKS

12 VCE-style practice questions

★ PRACTICE & SELF-MARK

Try each on paper first. Then reveal a weak answer, an annotated strong answer and the marking guide, and self-mark. Your total feeds the dock. "A stakeholder is someone involved. Two groups are the farmers and the environment." Definition is too thin; "the environment" is not a stakeholder group, and no priorities are given. "A stakeholder is any individual or group with an interest in, or affected by, an environmental decision. Two groups with differing priorities are Connawarra's irrigators, who prioritise secure water for farming, and the Boorla Traditional Owners, who prioritise cultural flows that flood the wetland." Crisp definition + two named groups + a contrasting priority each.

- 1 mark — correct definition of stakeholder (interest in / affected by the decision).
- 1 mark — two appropriately named groups with clearly differing priorities.

★ PRACTICE & SELF-MARK

"There are laws that protect the wetland so they have to give it water." Single vague point; no named framework, no mechanism. Caps at 1. "Regulatory frameworks both enable and require the flow. The Murray–Darling Basin Plan reserves a legal volume of environmental water, delivered by the Commonwealth Environmental Water Holder, which makes the release possible. The Ramsar Convention obliges Australia to maintain the wetland's ecological character, creating a duty to act. Meanwhile existing irrigation entitlements constrain how much can be reallocated, so the frameworks also set the upper limit." Three distinct, named frameworks each with a clear influence — enabling, requiring, constraining.

- 1 mark each (max 3) for a correctly named framework with its specific influence on the decision.
- Credit enabling, requiring or constraining roles.

★ PRACTICE & SELF-MARK

"An interconnection is when things connect and a tension is when they disagree. The farmers and the wetland have a tension." Defines loosely, gives only one example, no interconnection example, no clear contrast. "An interconnection is where two factors reinforce or depend on one another: for example, new sensor technology generates the scientific data the CMA uses to set watering targets — the two factors strengthen each other. A tension is where two factors pull in opposite directions: for example, irrigators' priority for maximum allocation conflicts with the Basin Plan's environmental-flow requirement, so a gain for one is a loss for the other. The key difference is that an interconnection aligns factors while a tension opposes them." Both terms defined, both illustrated with a distinct named example, and an explicit point of distinction.

- 1 mark — interconnection defined as reinforcing/dependent.
- 1 mark — valid interconnection example.
- 1 mark — tension defined as conflicting.
- 1 mark — valid tension example (the explicit contrast is what lifts it to full marks).

★ PRACTICE & SELF-MARK

"The bird data is bad so they should fix the swamp. The farmers don't want to pay more. There are rules about water quality." Spots three factors but doesn't name them, doesn't interpret the data conflict, and never links or weighs them. "Scientific data pulls two ways: twenty years of survey data (historical) shows a breeding collapse, justifying action, yet the recent wet winter (current) could be misinterpreted as recovery — a reason to apply the precautionary principle. Stakeholder priorities create a tension: oyster farmers prioritise low costs, raising a user-pays/equity issue against the swamp's ecological need. New technology (recycled-water piping) makes the intervention possible but introduces a cost. The regulatory framework (state water-quality standard) constrains the scheme, interconnecting with the data because approval depends on monitoring. A responsible decision weighs the long-term ecological evidence above the short-term wet-year signal while addressing who bears the cost." All four factors named, historical-vs-current data interpreted, an interconnection and a tension identified, and a weighted judgement.

- Up to 4 marks — each factor (S, R, D, T) correctly identified and applied to the stimulus.
- 1 mark — at least one explicit interconnection or tension between factors, or a reasoned weighting.
- Reward use of "historical and current" data interpretation.

★ PRACTICE & SELF-MARK

"All four factors matter. Stakeholders disagree, there are laws, data is collected and technology helps. The decision should be sustainable for the future." Mentions all four but with no examples, no interconnection/tension analysis and only a token nod to principles. Stalls in the middle band. "The four factors are deeply interwoven. Scientific data (D) — river red gum canopy-condition surveys and three failed breeding seasons — triggers the proposal, and is interconnected with regulatory frameworks (R): the Basin Plan and Ramsar both rely on that evidence to justify reserving environmental water. Stakeholder values (S) create tension: the Boorla Traditional Owners and the CMA (ecocentric) support the flood, while some irrigators (anthropocentric) fear reduced allocation — a tension within the single factor S. New technology (T) — soil-moisture sensors and drip irrigation — eases that S–R tension by cutting on-farm demand, though the cost raises a user-pays issue. Weighing these, the responsible decision favours the flood, justified by intergenerational equity and ecological integrity, while the precautionary principle supports acting despite data uncertainty. The decision is responsible because it is evidence-based, lawful, transparent about trade-offs and defensible against the sustainability principles." Every factor exemplified, both an interconnection and a tension analysed, value systems used, and three principles applied to reach a justified position.

- Up to 4 marks — each factor identified with a case-study example.
- 1 mark — at least one interconnection and one tension explicitly analysed.
- 1 mark — decision linked to named sustainability principles with a justified judgement.
- Band descriptor: 5–6 = integrated discussion weighing factors; 3–4 = factors explained but loosely linked; 1–2 = list with little analysis.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

A stakeholder is someone involved. Two groups are the farmers and the environment." Definition is too thin; "the environment" is not a stakeholder group, and no priorities are given.

✓ STRONG / MODEL ANSWER

A stakeholder is any individual or group with an interest in, or affected by, an environmental decision. Two groups with differing priorities are Connawarra's irrigators, who prioritise secure water for farming, and the Boorla Traditional Owners, who prioritise cultural flows that flood the wetland." Crisp definition + two named groups + a contrasting priority each.

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There are laws that protect the wetland so they have to give it water." Single vague point; no named framework, no mechanism. Caps at 1.

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